

OPERATING SYSTEM — Course Roadmap

Course Code: 23CM4114 | Credits: 3 | 4 Theory / Week | Dept. of CSE-AI&ML; , ANITS

Operating System Concepts

Text: Silberschatz, Galvin & Gagne — OS Concepts, 10th Ed., Wiley, 2018

UNIT I Intro to OS & Shell CO1	UNIT II Process Management CO2	UNIT III Sync & Deadlock CO3	UNIT IV Memory Management CO4	UNIT V File Systems & Protection CO5
Overview of OS	Process Concept & PCB	Critical Section Problem	Logical vs Physical Addr.	File Concepts & Access Methods
Types of OS	Process Scheduling	Peterson's Solution	Contiguous Allocation	Directory Structure
OS Structures	Operations on Process	Sync Hardware	Paging & TLBs	File System Implementation
OS Services	IPC	Semaphores	Segmentation	Disk Scheduling
System Calls	Client-Server Comm.	Classical Problems	Swapping	I/O Systems
Virtual Machines	Threads & Multithreading	Monitors	Virtual Memory	Goals of Protection
Shell Programming	CPU Scheduling Algorithms	Deadlock: Avoid, Detect & Recover	Page Replacement & Thrashing	Access Matrix

■ Unit I — Intro & Shell (10 Th)

■ Unit II — Process Mgmt (12 Th)

■ Unit III — Sync & Deadlock (12 Th)

■ Unit IV — Memory Mgmt (10 Th)

■ Unit V — File & Protection (10 Th)

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